2017

2. (a) (i) $\overline{A+B\bar{C}+D(E+\bar{F})}$

$= \overline{\overline{A+B\bar{C}} \cdot D(E+\bar{F})}$

$= (A+B\bar{C})\left(\bar{D}+\overline{(E+\bar{F})}\right)$

$= (A+B\bar{C})(\bar{D}+E+\bar{F})$

(ii) $\overline{\overline{ABC}+D+F}$

$= \overline{\overline{ABC}} \cdot \bar{D}\bar{F}$

$= ABC\bar{D}\bar{F}$

(b) $[AB(C+(BD)')+(AB)']C$

$= [AB(C+\bar{B}\bar{D})+\bar{A}\bar{B}]C$

$= [AB(C+\bar{B}+\bar{D})+\bar{A}+\bar{B}]C$

$= (ABC+AB\bar{B}+AB\bar{D}+\bar{A}+\bar{B})C$

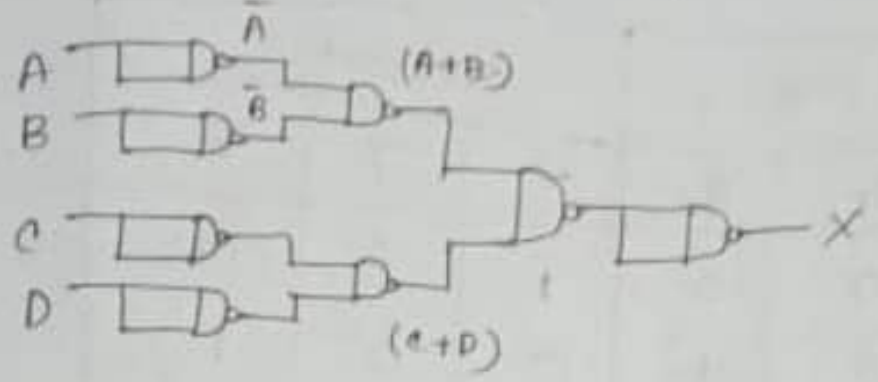
$= ABCC+ABC\bar{D}+\bar{A}C+\bar{B}C$

$= ABC+AB\bar{D}+\bar{A}C+\bar{B}C$

$= C(AB$

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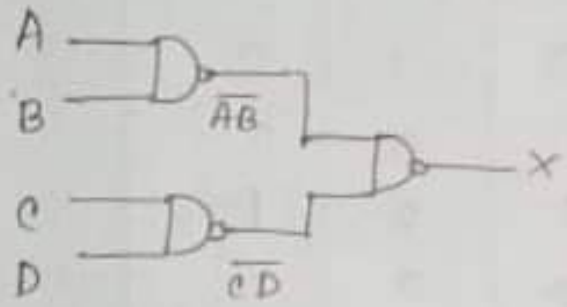
(c) $X = (A+B)(C+D)$ using NAND Gate



$\overline{\bar{A}\cdot\bar{B}}$

$= \bar{\bar{A}} + \bar{\bar{B}}$

$X = AB + CD$ using NAND Gate



$\overline{\overline{AB}\cdot\overline{CD}}$

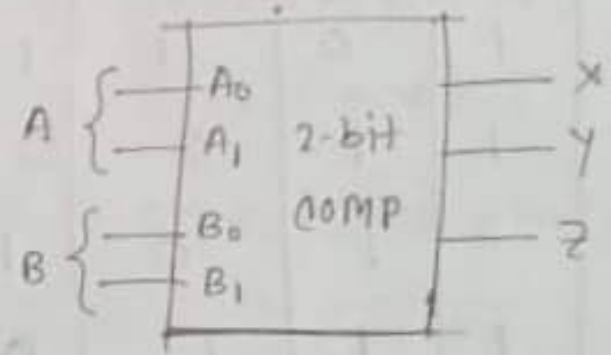
$= AB + CD$

3.

2-bit comparator

~~truth table~~

block-diagram



if $A = B$, $X = 1, Y = 0, Z = 0$

if $A > B$, $X = 0, Y = 1, Z = 0$

if $A < B$, $X = 0, Y = 0, Z = 1$

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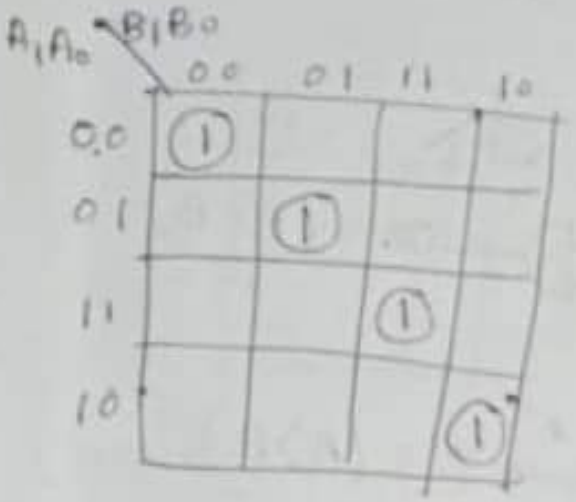
truth table

Input				output		
A_1	A_0	B_1	B_0	X	Y	Z
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

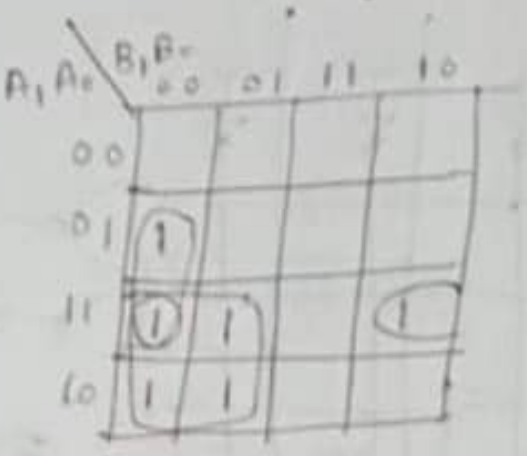
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logical expression

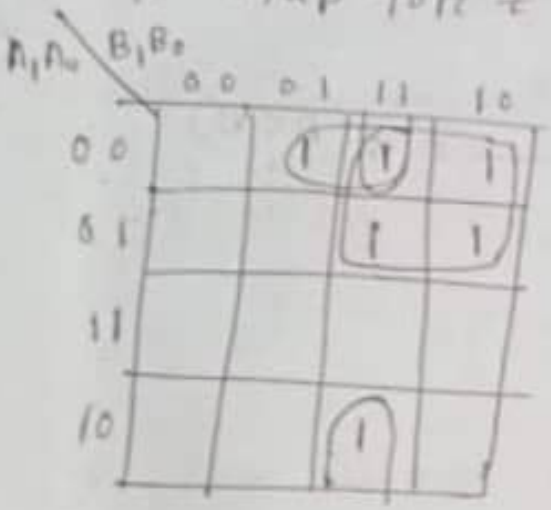
K-map for X,



K map for Y.



K map for Z



$X = \bar{A_1} A_0 \bar{B_1} B_0 + \bar{A_1} A_0 \bar{B_1} B_0 +$

$A_1 \bar{A_0} B_1 \bar{B_0} + A_1 A_0 B_1 B_0$

$Z = \bar{A_1} \bar{B_1} (\bar{A_0} \bar{B_0} + A_0 B_0) +$

$A_1 B_1 (\bar{A_0} \bar{B_0} + A_0 B_0)$

$= (\bar{A_1} \bar{B_1} + A_1 B_1)(\bar{A_0} \bar{B_0} + A_0 B_0)$

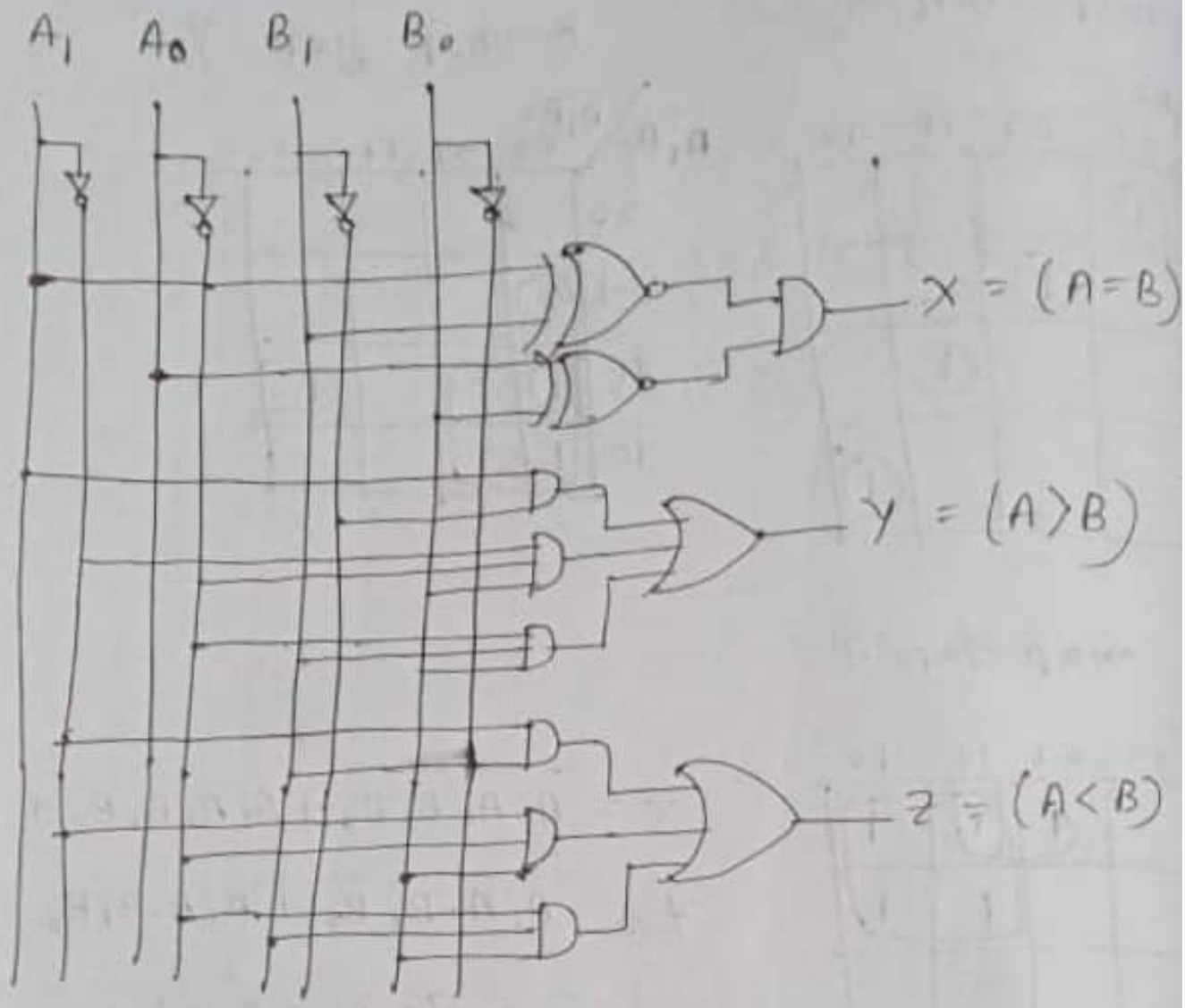
$= (\overline{A_1 \oplus B_1})(\overline{A_0 \oplus B_0})$

$Y = A_1 \bar{B_1} + A_0 \bar{B_1} \bar{B_0} + A_1 A_0 \bar{B_0}$

$Z = \bar{A_1} B_1 + \bar{A_1} \bar{A_0} B_0 + \bar{A_0} B_1 B_0$

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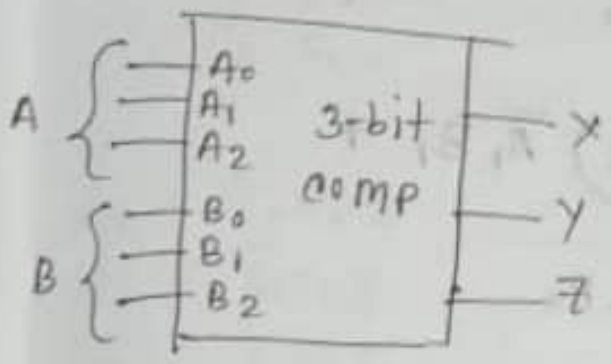
circuit



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3 bit comparator

block diagram



condition

1. if $A_2 = B_2$, $A_1 = B_0$, $A_0 = B_0$ then $A = B$

2. ~~if the output is $A < B$ in cases~~

$$x = (A = B) = (\bar{A}_2\bar{B}_2 + A_2B_2)(\bar{A}_1\bar{B}_1 + A_1B_1)(\bar{A}_0\bar{B}_0 + A_0B_0)$$

$$= \overline{(A_2 \oplus B_2)}\ \overline{(A_1 \oplus B_1)}\ \overline{(A_0 \oplus B_0)}$$

2. the output is $A > B$ in cases of

$A_2 > B_2$

$A_2 = B_2$ and $A_1 > B_1$

$A_2 = B_2$, $A_1 = B_1$ and $A_0 > B_0$

$$y = A_2\bar{B}_2 + \overline{(A_2 \oplus B_2)}\,A_1\bar{B}_1 + \overline{(A_2 \oplus B_2)}\ \overline{(A_1 \oplus B_1)}\ A_0\bar{B}_0$$

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3. the output is $A<B$ if

$A_2 < B_2$

$A_2 = B_2$ and $A_1 < B_1$

$A_2 = B_2$, $A_1 = B_1$ and $A_0 < B_0$

$$Z = (A<B) = \overline{A_2} B_2 + (\overline{A_2 \oplus B_2})\, \overline{A_1} B_1 + (\overline{A_2 \oplus B_2})(\overline{A_1 \oplus B_1})\, \overline{A_0} B_0$$

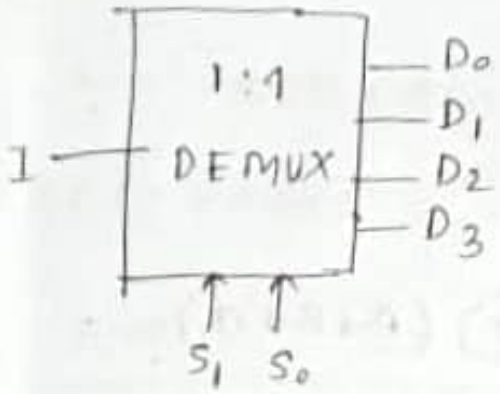
(b)

Demultiplexer: It is actually the reverse funation of multiplexing. It take digital information from one input line and distributes it to a given number of output lines so it is known as data distributor.

1:4 DEMUX

it has 1 input line, 4 output lines and 2 select lines.

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truth table

S_1	S_0	D_0	D_1	D_2	D_3
0	0	I	0	0	0
0	1	0	I	0	0
1	0	0	0	I	0
1	1	0	0	0	I

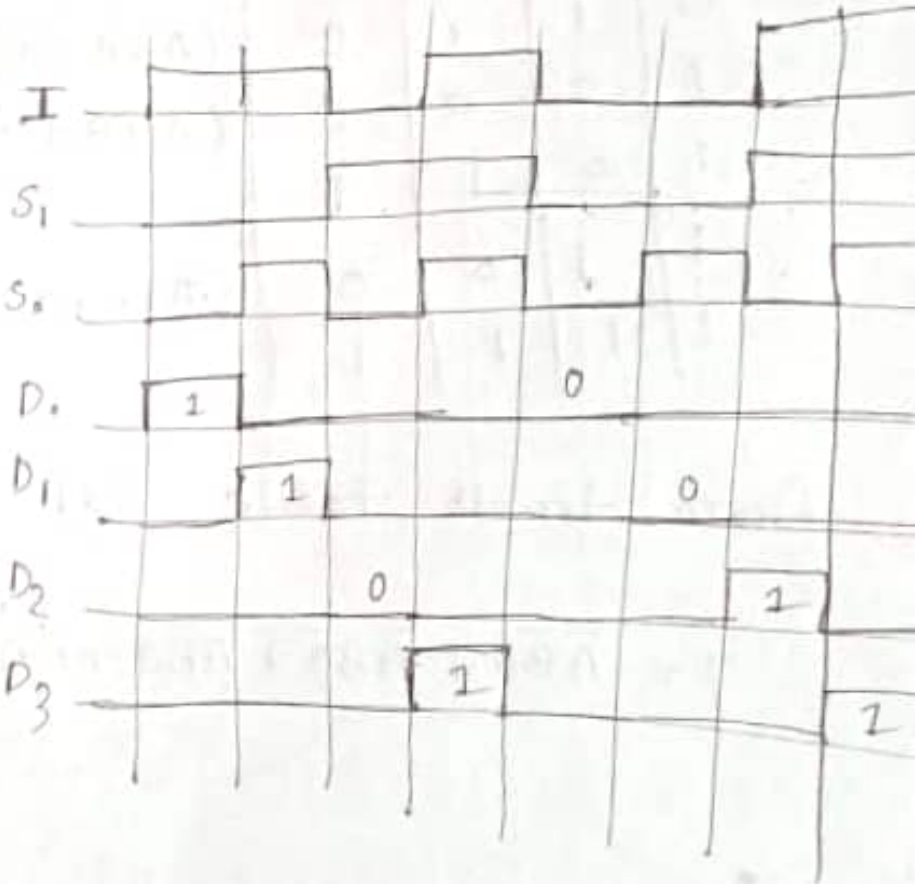
$D_0 = \bar{S}_1 \bar{S}_0 I$

$D_1 = \bar{S}_1 S_0 I$

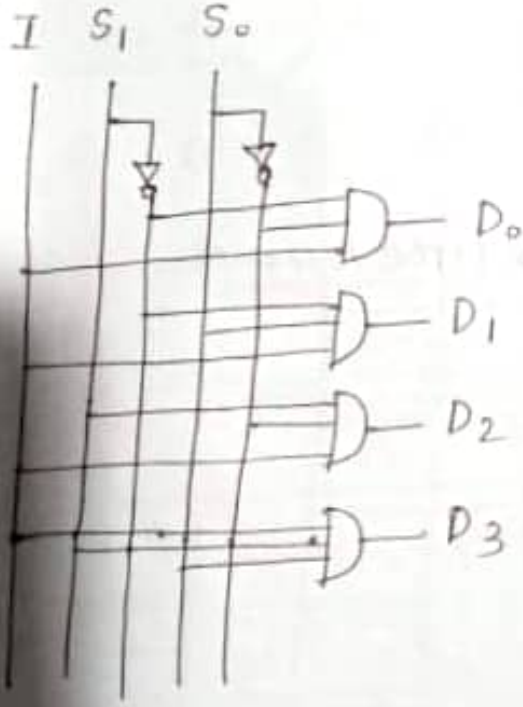
$D_2 = S_1 \bar{S}_0 I$

$D_3 = S_1 S_0 I$

timing diagram



circuit



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4.

(a) Suppose,

$Y = (\bar{A}+B+C)(\bar{A}+\bar{B}+C)(A+\bar{B}+\bar{C})(A+B+C)$

A	B	C	Y	terms
0	0	0	0	$(A+B+C)$
0	0	1	1	
0	1	0	1	
0	1	1	0	$(A+\bar{B}+\bar{C})$
1	0	0	0	$(\bar{A}+B+C)$
1	0	1	1	
1	1	0	0	$(\bar{A}+\bar{B}+C)$
1	1	1	1	

from truth table, SOP expression

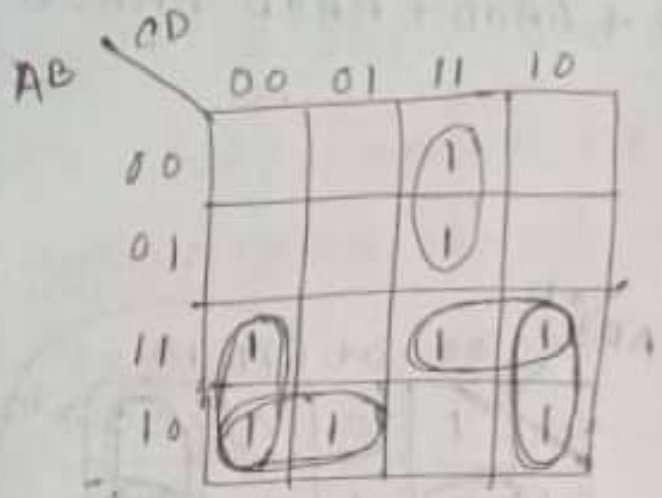
$Y = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + ABC$

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(b)

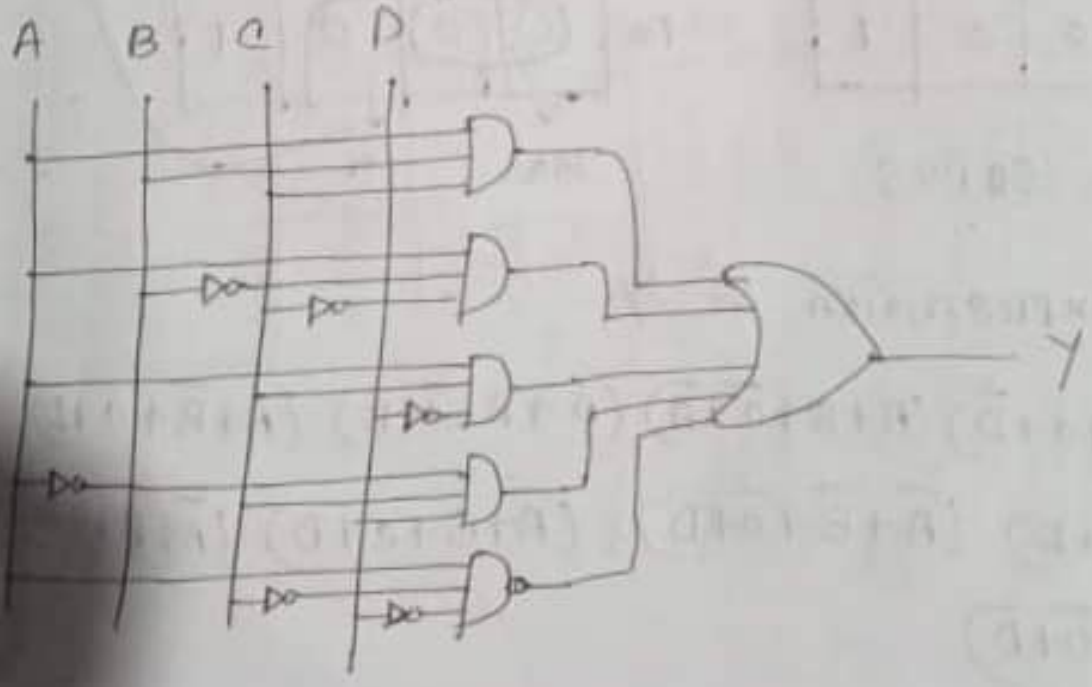
$Y = \bar{A}BC\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + ABC\bar{D} + AB\bar{C}\bar{D} + ABCD$

k-map for Y



$Y = \bar{A}CD + A\bar{C}\bar{D} + AB\bar{C} + ABC + AC\bar{D}$

circuit



Scanned with CamScanner

(c)

$Y = \bar{A}B\bar{C}D + AB\bar{C}\bar{D} + ABCD + A\bar{B}C\bar{D} + \bar{A}BCD$

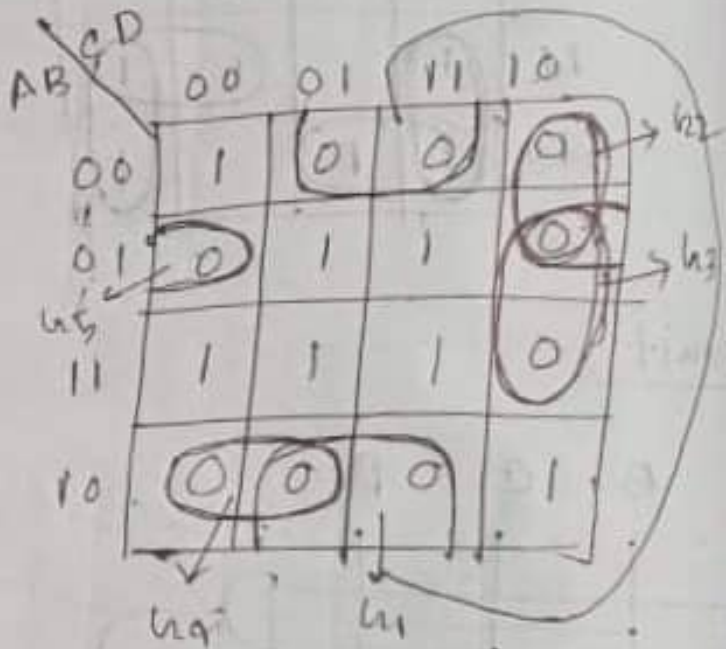
$+ \bar{A}B\bar{C}D + AB\bar{C}\bar{D}$

$= \bar{A}BCD + AB\bar{C}D + AB\bar{C}\bar{D} + ABCD + A\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$

$+ \bar{A}B\bar{C}D$

K-map for Y,

AB \ CD	00	01	11	10
00	1	0	0	0
01	0	1	1	0
11	1	1	1	0
10	0	0	0	1



for SOPOS

SPOS expression of Y,

$Y = (A+B+C+\bar{D})(A+B+\bar{C}+\bar{D})(A+B+\bar{C}+D)(A+\bar{B}+C+D)$

$(A+\bar{B}+\bar{C}+D)(\bar{A}+\bar{B}+\bar{C}+D)(\bar{A}+B+C+D)(\bar{A}+B+C+\bar{D})$

$(\bar{A}+B+\bar{C}+\bar{D})$

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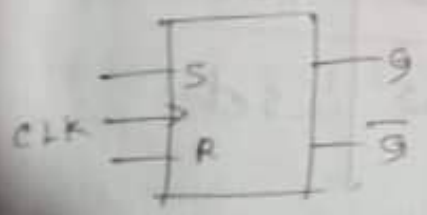
minimum POS,

$$Y = (B+\bar{D})(A+\bar{C}+D)(\bar{B}+\bar{C}+D)(\bar{A}+B+C)(A+\bar{B}+D)$$

5.

flip-flop: flip-flops are synchronous bistable devices, also known as bistable multivibrators where synchronous means the output changes state only at a specified point (leading or trailing edge) on the triggering input called the clock (CLK).

SR flip flop :

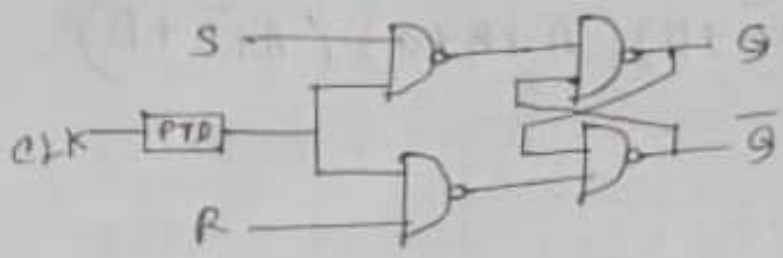


truth table

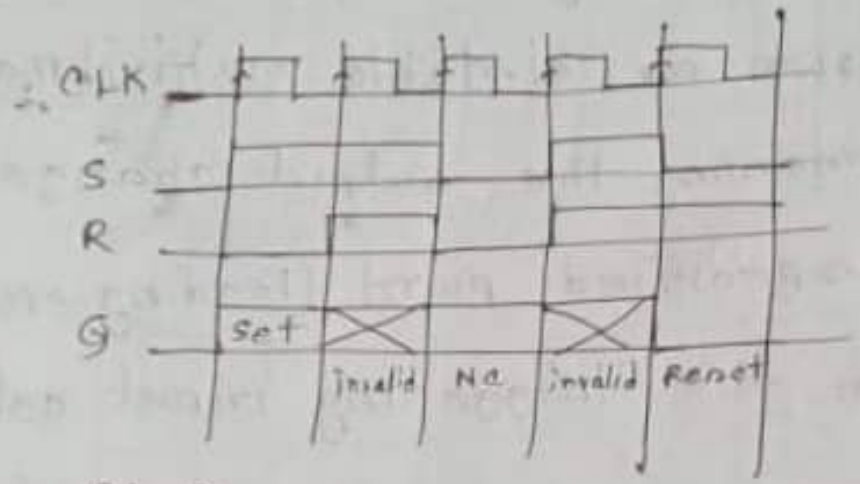
CLK	S	R	Q^t	Q^{t+1}	comment
↑	0	0	0	0	NC
			1	1	
↑	0	1	0	0	Reset
			1	0	
↑	1	0	0	1	Set
			1	1	
↑	1	1	0	x	invalid
			1	x	

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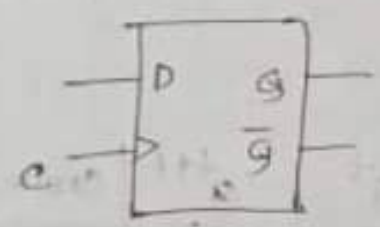
circuit



timing diagram

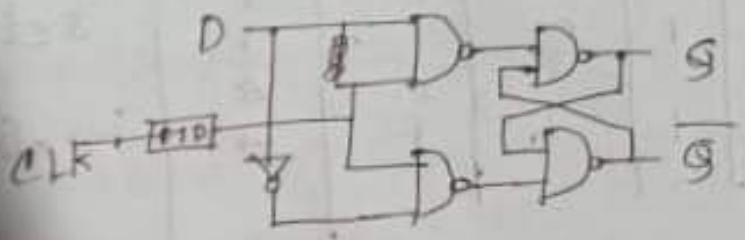


D-flip-flop



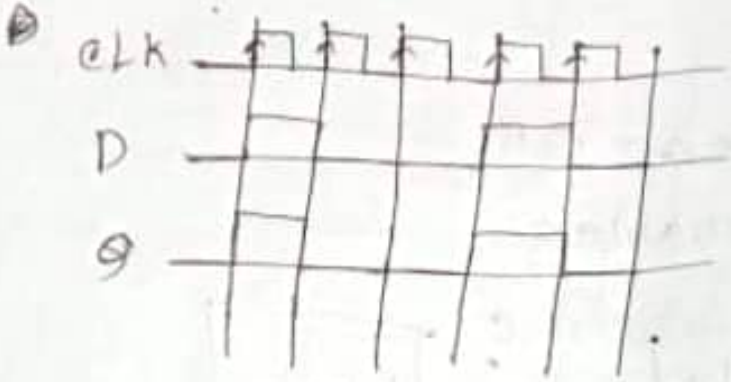
truth table

CLK	D	Q	$\bar{Q}$	comment
↑	0	0	1	Reset
↑	1	1	0	set



Scanned with CamScanner

timing diagram



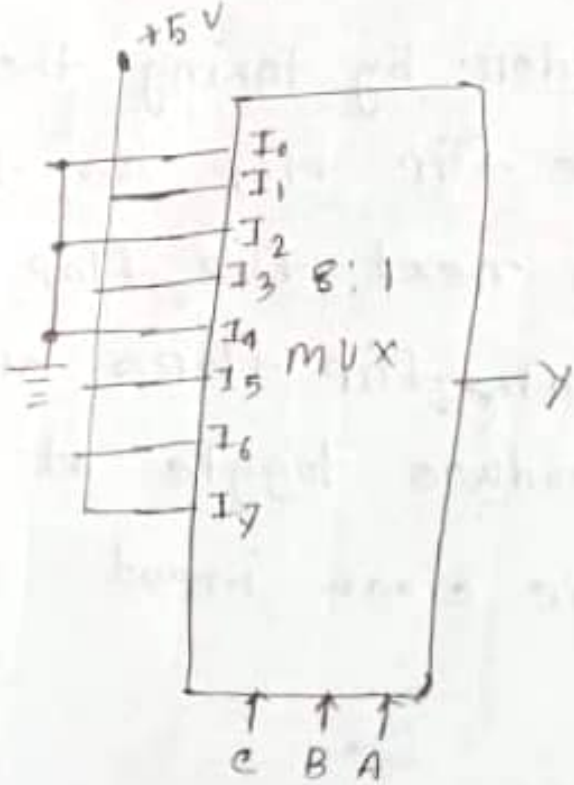
(b) $Y = \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}C + ABC + AB\bar{C}$

dec	A	B	C	Y
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

using 8:1 MUX

$2^3 \rightarrow 8$

3 select lines — A, B, C



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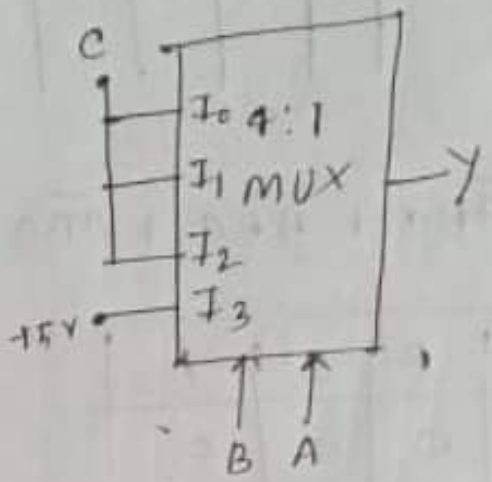
using 4:1 MUX

$2^2 \rightarrow 4$

2 select lines = A, B

c → input variables

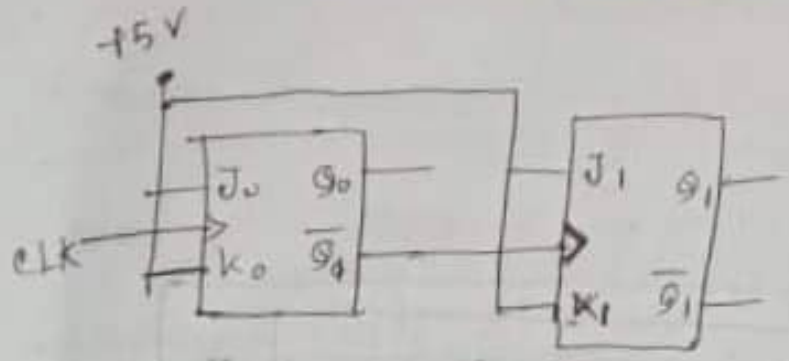
	I_0	I_1	I_2	I_3
$\bar{c}$	0	2	4	(6)
c	(1)	(3)	(5)	(7)
	c	c	c	1



6.(a)

J-K flip flop can be used as frequency divider by taking the output of one ~~one~~ flip flop as the clock input to the next flip flop. J and K input of each flip flops are set to 1 to produce toggle at a each cycle of the clock input.

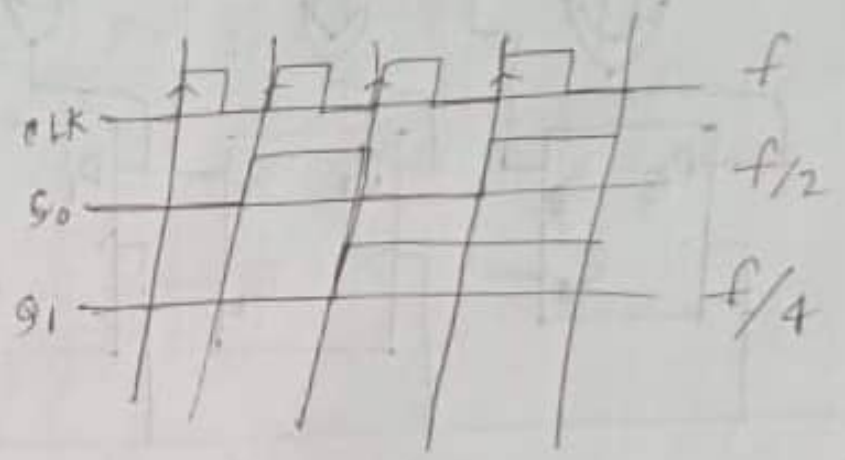
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State sequence
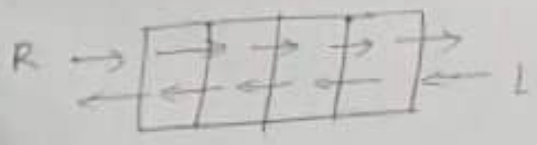
clk	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1

timing diagram



(b)

bidirectional shift registers: It is one in which the data bits can be shifted from left to right or right to left

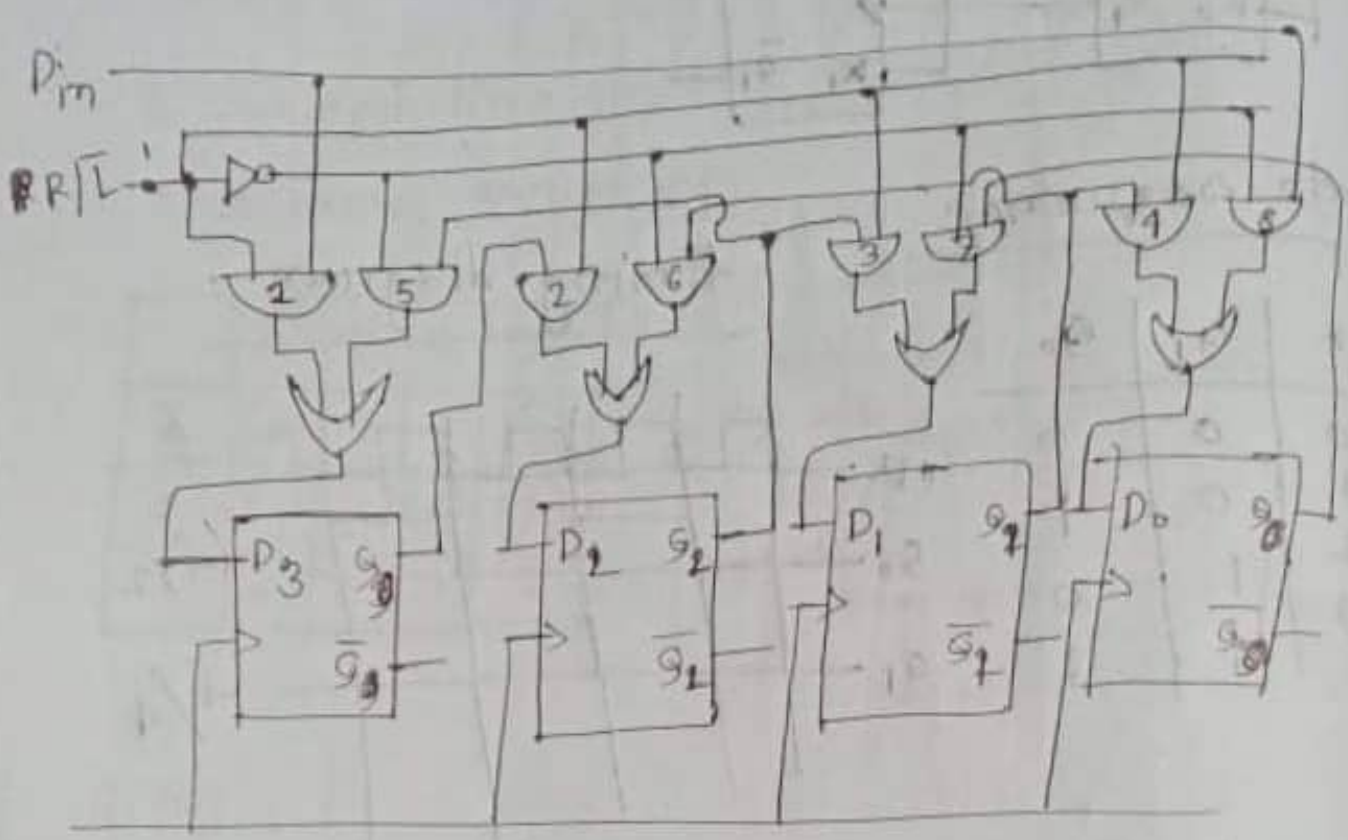


$R/\overline{L} = 1$ → Right shift

$R/\overline{L} = 0$ → Left shift

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circuit



$R/\bar{L} = 1 \rightarrow$ 1, 2, 3, 4 gates ~~are~~ ~~ables~~ will be enabled and 5, 6, 7, 8 gates will be disabled.

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timing diagram

~~ex: 1101~~

~~Suppose that, Din = 0~~

ex: 1011

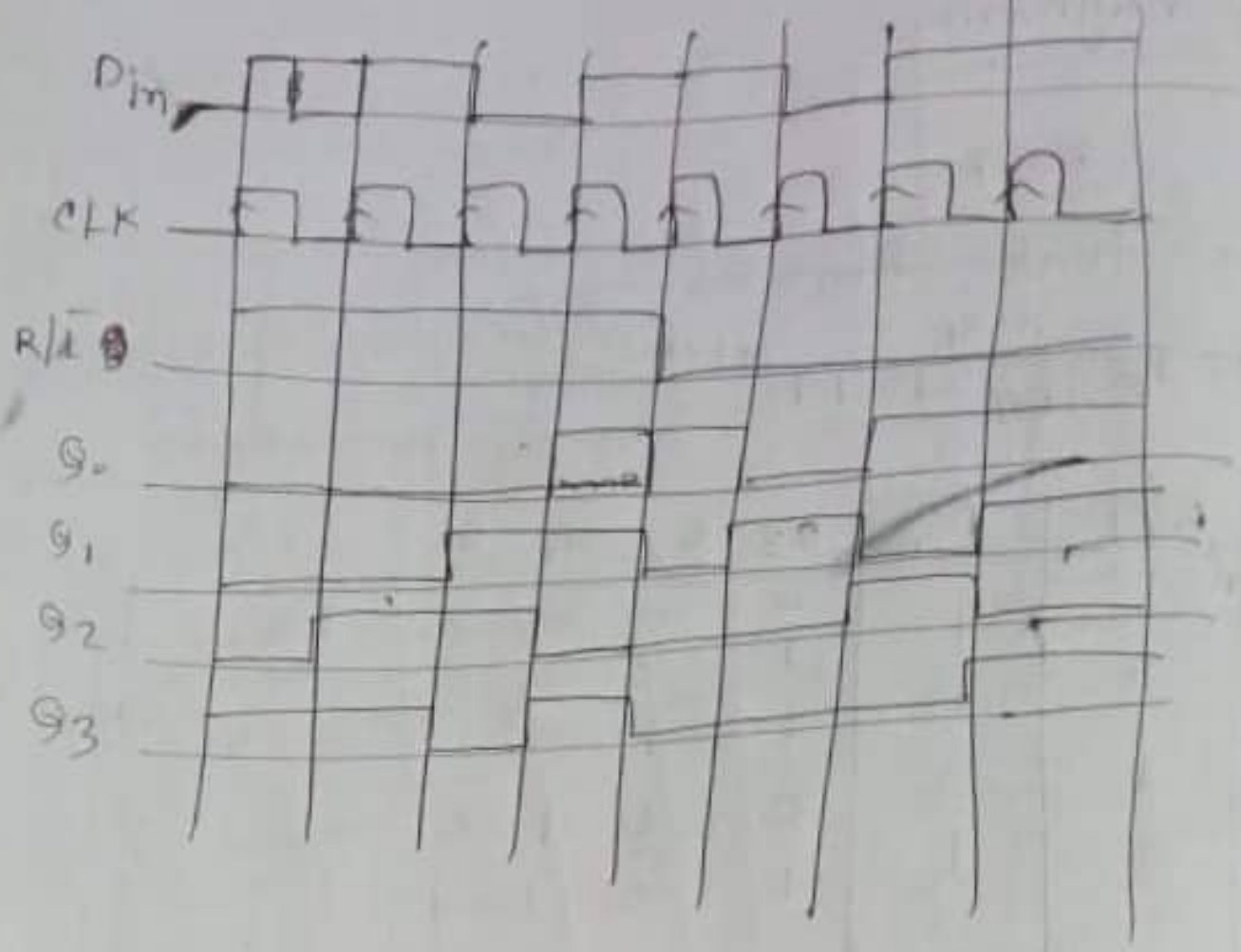
MSB → LSB

CLK	R/$\bar{L}$	Din	Q_3	Q_2	Q_1	Q_0
0						
1	1	1	0	0	0	0
2	1	1	1	0	0	0
3	1	0	1	1	0	0
4	1	1	0	1	1	0
			1	0	1	1

CLK	R/$\bar{L}$	Din	Q_0	Q_1	Q_2	Q_3
5	0	1	1	0	0	1
6	0	0	1	1	0	0
7	0	1	1	0	1	0
8	0	1	1	1	0	1

timing diagram

Scanned with CamScanner

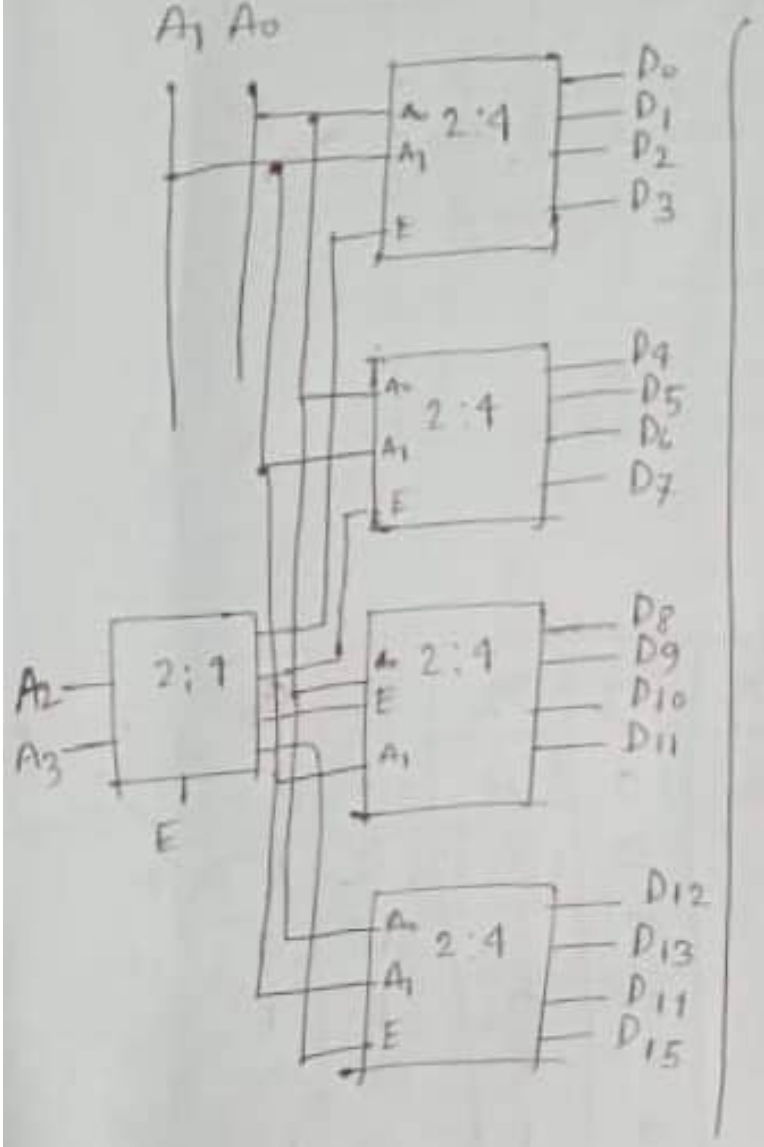


CLK	Din	R/$\bar{L}$	Q_0	Q_1	Q_2	Q_3
0	0	0	1	1	0	1
1	0	1	0	1	1	0
2	0	1	0	0	1	1
3	0	0	0	1	1	0
4	0	0	1	1	0	0
5	0	0	1	0	0	0
6	0	1	0	1	0	0
7	0	1	0	0	1	0
8	0	0	0	1	0	0
9	0	0	1	0	0	0

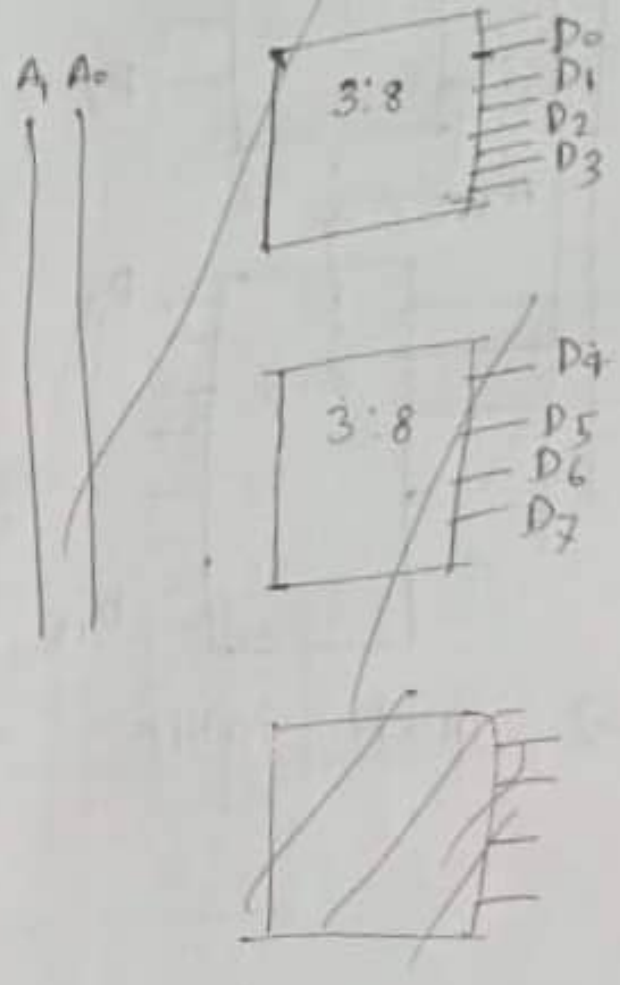
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7.(a)

4:16 decoder using 2:4 decoder

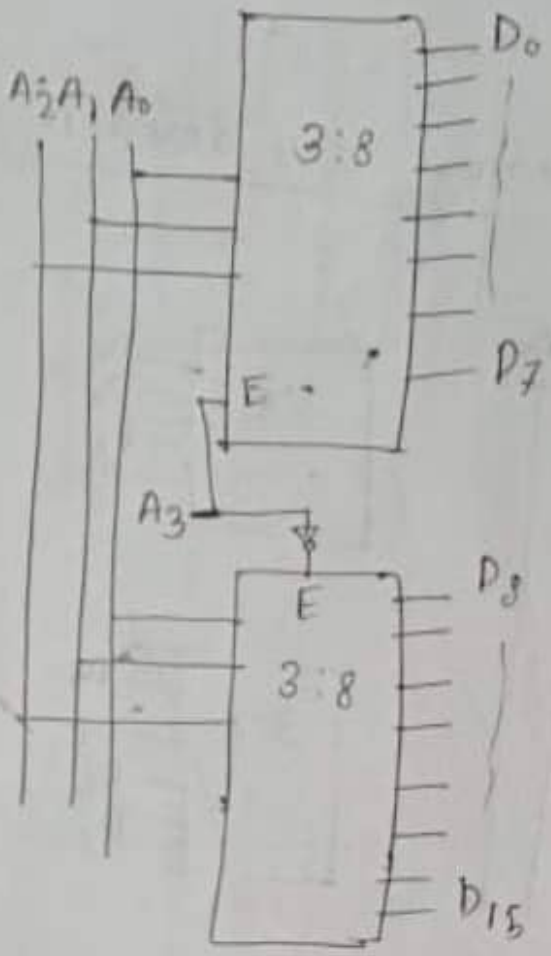


using 3:8 decoder



Scanned with CamScanner

3:8 decoder



(b) truth table

clk	S_3	S_2	S_1	S_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	0	0	0	0

Scanned with CamScanner

flip-flop transition table

out input		in output	
Q^t	Q^{t+1}	J	K
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

k-map

for J_0,

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	1	*	*	1
01	1	*	*	1
11	*	*	*	*
10	1	*	*	0

$J_0 = \bar{Q}_1 + \bar{Q}_3$

for K_0,

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	*	1	1	*
01	*	1	1	*
11	*	*	*	*
10	*	1	*	*

$K_0 = 1$

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for J_2,

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	1	*	*
01	0	1	*	*
11	*	*	*	*
10	0	1	*	*

$J_2 = Q_0$

for K_2,

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	*	*	1	0
01	*	*	1	0
11	*	*	*	*
10	*	*	*	1

$K_2 = Q_3 + Q_0$

for J_2,

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	*	*	*	*
11	*	*	*	*
10	0	0	*	0

$J_2 = Q_1Q_0$

K_2

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	*	*	*	*
01	0	0	1	0
11	*	*	*	*
10	*	*	*	*

$K_2 = Q_1Q_0$

J_3

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	*	*	*	*
10	*	*	*	*

K_3

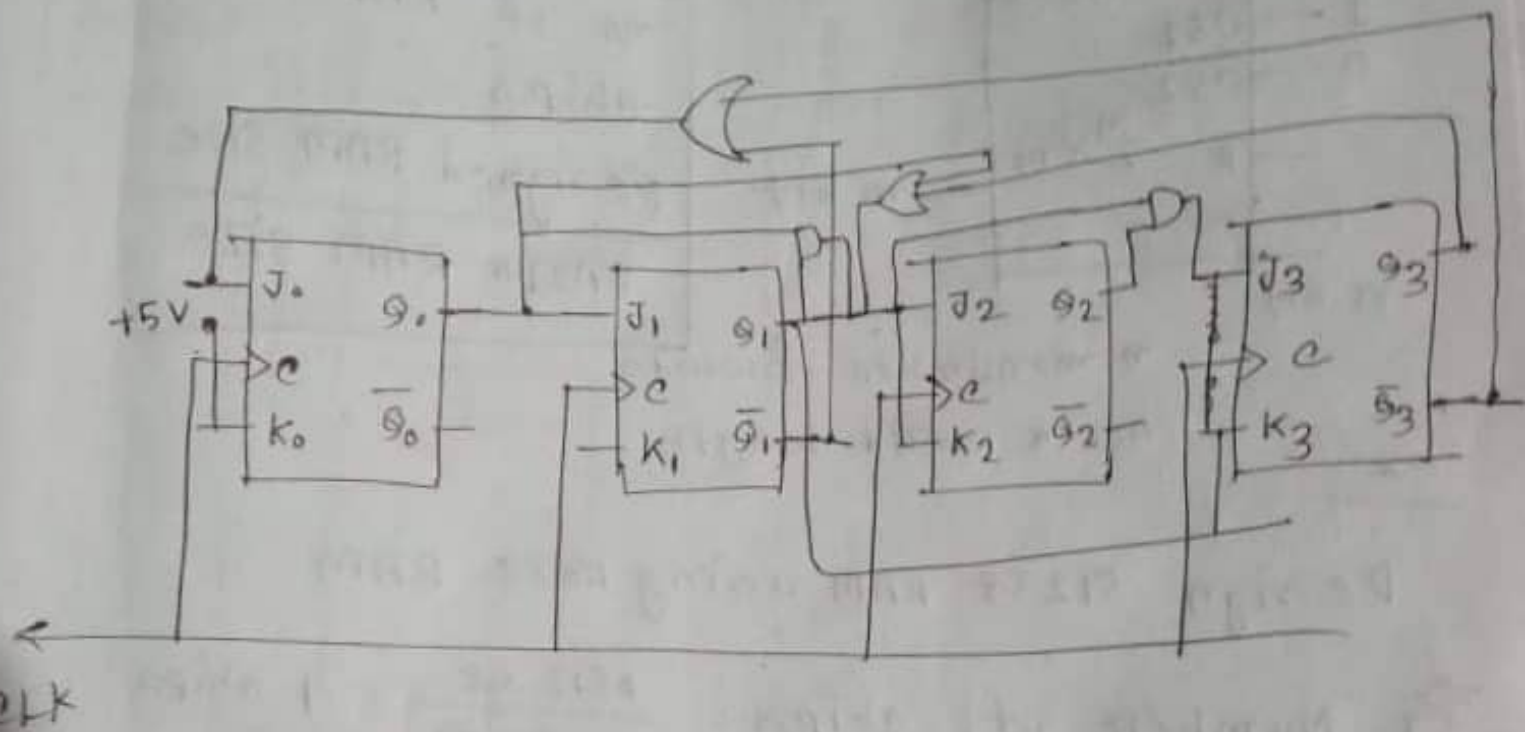
Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	*	*	*	*
01	*	*	*	*
11	*	*	*	*
10	0	0	*	1

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$J_3 = Q_0 Q_1 Q_2$

$K_3 = Q_1$

Circuit



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8. (a)

32×8 RAM using 8×1 RAM chip

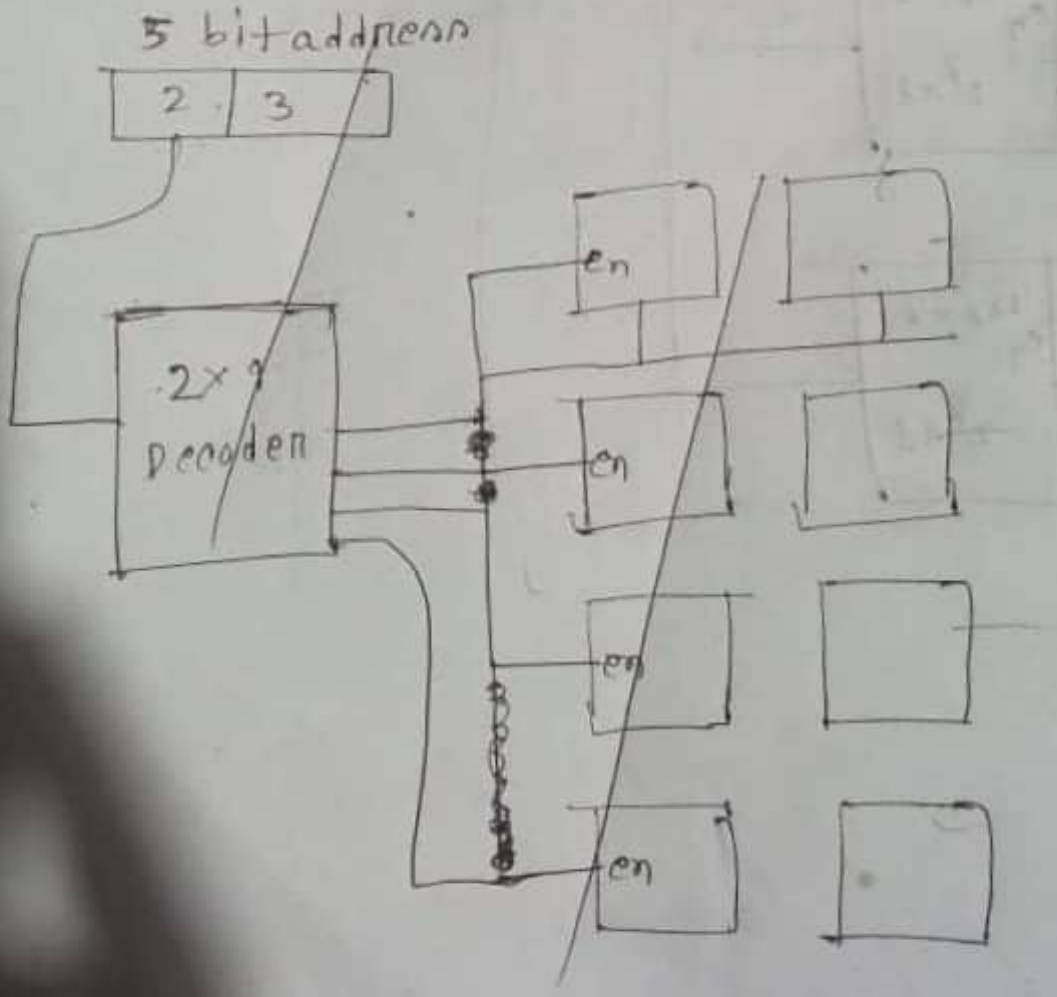
⇒

$$\text{no. of chips} = \frac{32 \times 8}{8 \times 1} = 8$$

bit address = $32 \rightarrow 2^5$ 5 bit

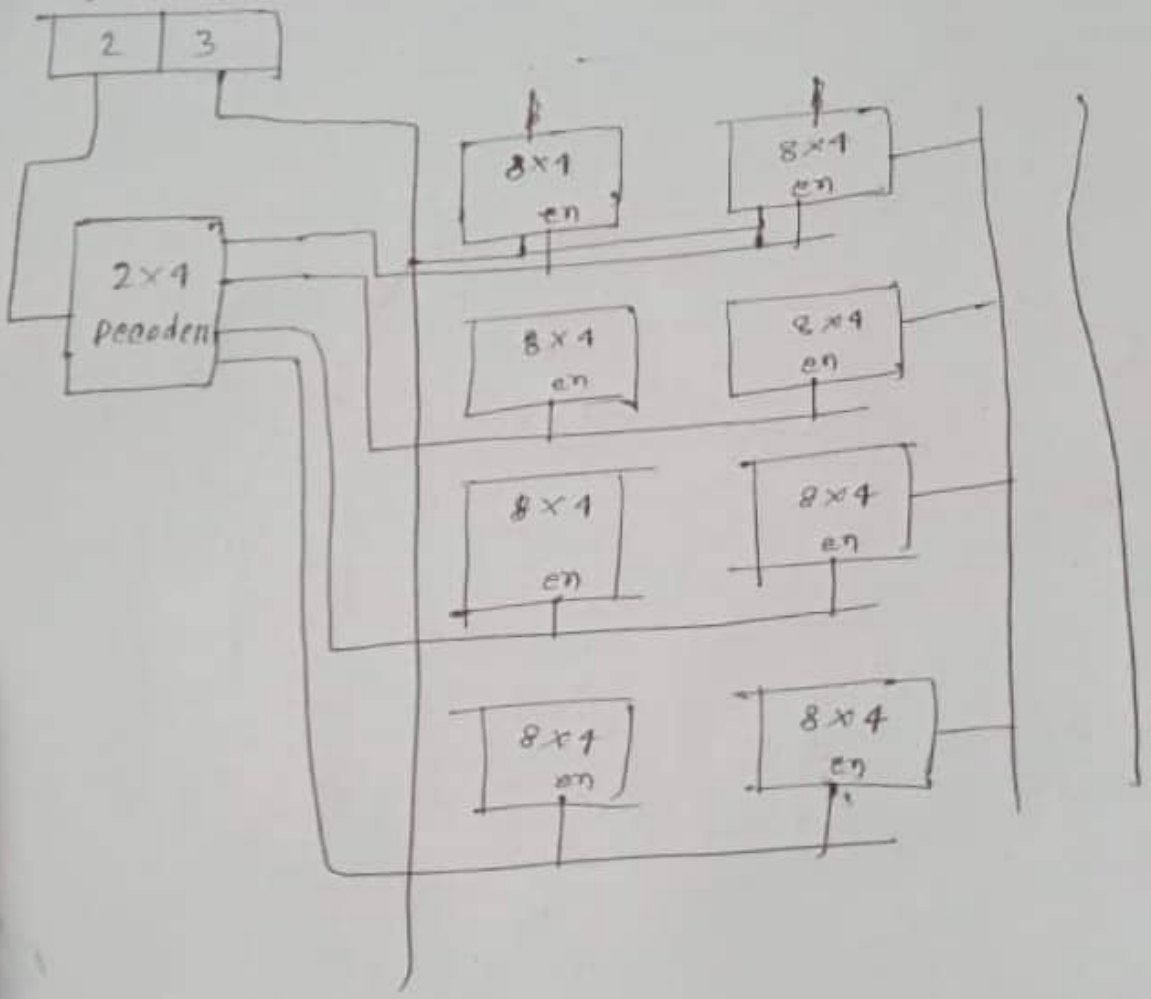
$$\text{Decoder size} = \frac{32}{8} = 4$$

2×1 decoder



Scanned with CamScanner

32×8 RAM using 8×4



Scanned with CamScanner